

Data Blog (/)

Data Science, Machine Learning and Statistics, implemented in Python

Support Vector Machine: calculate coefficients manually

Xavier Bourret Sicotte

Mon 25 June 2018

Category: Machine Learning (/category/machine-learning.html)

Support Vector Machines

In this first notebook on the topic of Support Vector Machines, we will explore the intuition behind the weights and coefficients by solving a simple SVM problem by hand.

The code in this notebook served to produce the following stats.stackexchange posts:

- <https://stats.stackexchange.com/questions/39243/how-does-one-interpret-svm-feature-weights/355043#355043> (<https://stats.stackexchange.com/questions/39243/how-does-one-interpret-svm-feature-weights/355043#355043>)
- <https://stats.stackexchange.com/questions/353552/support-vector-machine-calculate-w-by-hand/353705#353705> (<https://stats.stackexchange.com/questions/353552/support-vector-machine-calculate-w-by-hand/353705#353705>)

Source:

- https://ai6034.mit.edu/wiki/images/SVM_and_Boosting.pdf
(https://ai6034.mit.edu/wiki/images/SVM_and_Boosting.pdf)

Libraries and utility functions

```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt
import seaborn as sns
sns.set()
```

```
def geo_margin(y,w,b,x):  
    return y*(w / np.linalg.norm(w)).T @ x + (b / np.linalg.norm(w)).flatten()  
  
def functional_margin(y,w,b,x):  
    return y*(w.T @ x + b).flatten()  
  
def f(x, w, b, c=0):  
    # given x1, return x2 such that [x1,x2] are on the line w.x + b = c  
    return (-w[0] * x - b + c) / w[1]
```

Dataset and inspection

```
#Data set
x_neg = np.array([[3,4],[1,4],[2,3]])
y_neg = np.array([-1,-1,-1])
x_pos = np.array([[6,-1],[7,-1],[5,-3]])
y_pos = np.array([1,1,1])
x1 = np.linspace(-10,10)
x = np.vstack((np.linspace(-10,10),np.linspace(-10,10)))

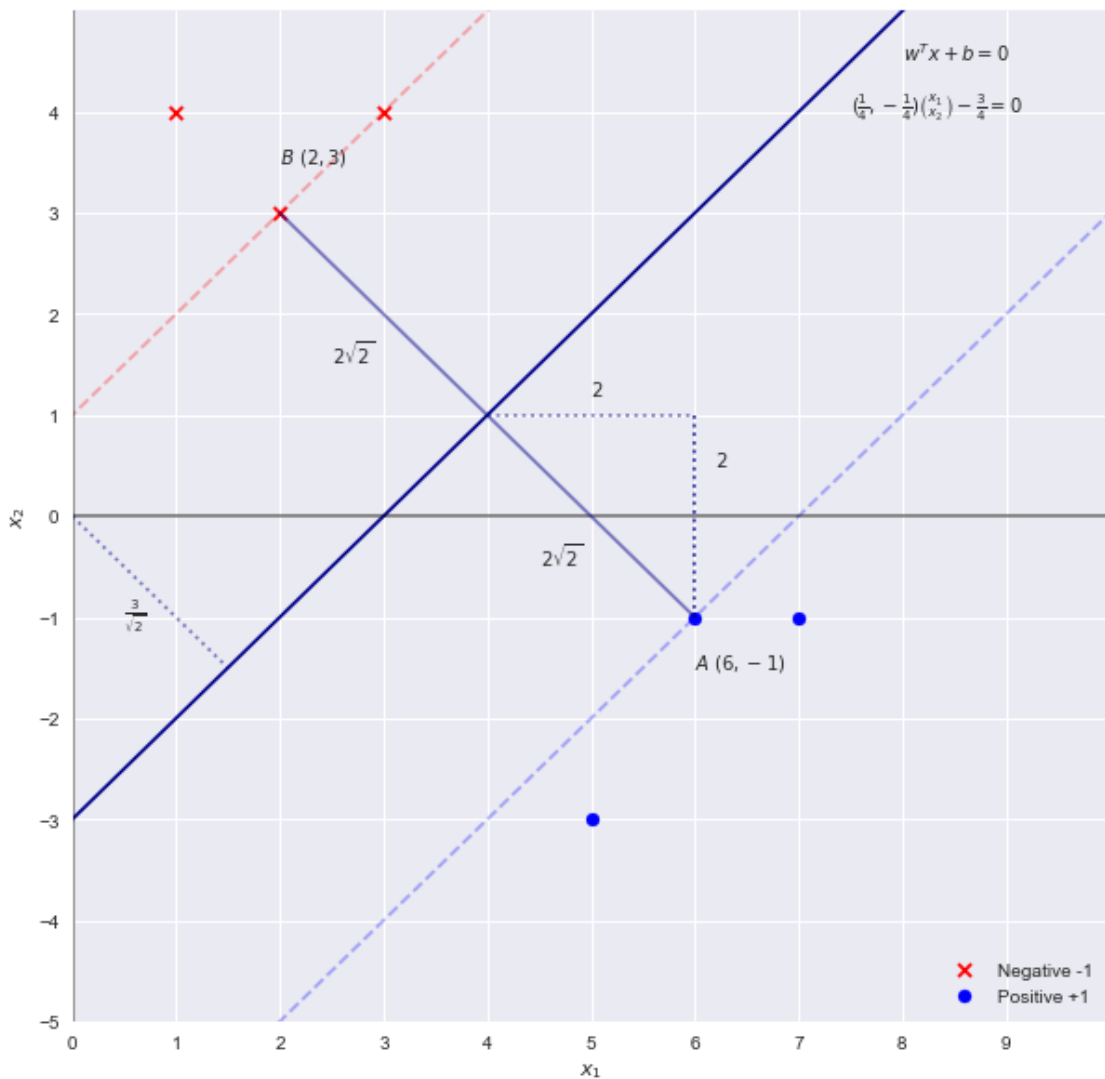
#Parameters guessed by inspection
w = np.array([1,-1]).reshape(-1,1)
b = -3

#Plot
fig = plt.figure(figsize = (10,10))
plt.scatter(x_neg[:,0], x_neg[:,1], marker = 'x', color = 'r', label = 'Negative -1')
plt.scatter(x_pos[:,0], x_pos[:,1], marker = 'o', color = 'b',label = 'Positive +1')
plt.plot(x1, x1 - 3, color = 'darkblue')
plt.plot(x1, x1 - 7, linestyle = '--', alpha = .3, color = 'b')
plt.plot(x1, x1 + 1, linestyle = '--', alpha = .3, color = 'r')
plt.xlim(0,10)
plt.ylim(-5,5)
plt.xticks(np.arange(0, 10, step=1))
plt.yticks(np.arange(-5, 5, step=1))

#Lines
plt.axvline(0, color = 'black', alpha = .5)
plt.axhline(0,color = 'black', alpha = .5)
plt.plot([2,6],[3,-1], linestyle = '-', color = 'darkblue', alpha = .5 )
plt.plot([4,6],[1,1],[6,6],[1,-1], linestyle = ':', color = 'darkblue', alpha = .5 )
plt.plot([0,1.5],[0,-1.5],[6,6],[1,-1], linestyle = ':', color = 'darkblue', alpha = .5 )

#Annotations
plt.annotate(s = '$A \ (6,-1)$', xy = (5,-1), xytext = (6,-1.5))
plt.annotate(s = '$B \ (2,3)$', xy = (2,3), xytext = (2,3.5))#, arrowprops = {'width':.2,
'headwidth':8})
plt.annotate(s = '$2$', xy = (5,1.2), xytext = (5,1.2) )
plt.annotate(s = '$2$', xy = (6.2,.5), xytext = (6.2,.5))
plt.annotate(s = '$2\sqrt{2}$', xy = (4.5,-.5), xytext = (4.5,-.5))
plt.annotate(s = '$2\sqrt{2}$', xy = (2.5,1.5), xytext = (2.5,1.5))
plt.annotate(s = '$w^Tx + b = 0$', xy = (8,4.5), xytext = (8,4.5))
plt.annotate(s = '$(\frac{1}{4},-\frac{1}{4}) \ \text{binom}\{x_1\}\{x_2\}- \ \frac{3}{4} = 0$', xy =
(7.5,4), xytext = (7.5,4))
plt.annotate(s = '$\frac{3}{\sqrt{2}}$', xy = (.5,-1), xytext = (.5,-1))

#Labels and show
plt.xlabel('$x_1$')
plt.ylabel('$x_2$')
plt.legend(loc = 'lower right')
plt.show()
```



Solving the SVM problem by inspection

By inspection we can see that the boundary decision line is the function $x_2 = x_1 - 3$. Using the formula $w^T x + b = 0$ we can obtain a first guess of the parameters as

$$w = [1, -1] \quad b = -3$$

Using these values we would obtain the following *width* between the support vectors: $\frac{2}{\sqrt{2}} = \sqrt{2}$. Again by inspection we see that the width between the support vectors is in fact of length $4\sqrt{2}$ meaning that these values are incorrect.

Recall that scaling the boundary by a factor of c does not change the boundary line, hence we can generalize the equation as

$$cx_1 - cx_2 - 3c = 0$$

$$w = [c, -c] \quad b = -3c$$

Plugging back into the equation for the width we get

$$\begin{aligned} \frac{2}{\|w\|} &= 4\sqrt{2} \\ \frac{2}{\sqrt{2}c} &= 4\sqrt{2} \\ c &= \frac{1}{4} \end{aligned}$$

Hence the parameters are in fact

$$w = [\frac{1}{4}, -\frac{1}{4}] \quad b = -\frac{3}{4}$$

To find the values of α_i we can use the following two constraints which come from the dual problem:

$$w = \sum_i^m \alpha_i y^{(i)} x^{(i)}$$

$$\sum_i^m \alpha_i y^{(i)} = 0$$

And using the fact that $\alpha_i \geq 0$ for support vectors only (i.e. 3 vectors in this case) we obtain the system of simultaneous linear equations:

$$\begin{bmatrix} 6\alpha_1 - 2\alpha_2 - 3\alpha_3 \\ -1\alpha_1 - 3\alpha_2 - 4\alpha_3 \\ 1\alpha_1 - 2\alpha_2 - 1\alpha_3 \end{bmatrix} = \begin{bmatrix} 1/4 \\ -1/4 \\ 0 \end{bmatrix}$$
$$\alpha = \begin{bmatrix} 1/16 \\ 1/16 \\ 0 \end{bmatrix}$$

Checking the results using Sklearn

```
from sklearn.svm import SVC

X = np.array([[3,4],[1,4],[2,3],[6,-1],[7,-1],[5,-3]] )
y = np.array([-1,-1, -1, 1, 1 , 1 ])

clf = SVC(C = 1e5, kernel = 'linear')
clf.fit(X, y)

print('w = ',clf.coef_)
print('b = ',clf.intercept_)
print('Indices of support vectors = ', clf.support_)
print('Support vectors = ', clf.support_vectors_)
print('Number of support vectors for each class = ', clf.n_support_)
print('Coefficients of the support vector in the decision function = ', np.abs(clf.dual_coef_))
```

```
w = [[ 0.25 -0.25]]
b = [-0.75]
Indices of support vectors = [2 3]
Support vectors = [[ 2.  3.]
 [ 6. -1.]]
Number of support vectors for each class = [1 1]
Coefficients of the support vector in the decision function = [[0.0625
 0.0625]]
```

Comments

ALSO ON [XAVIERBOURRETSICOTTE.GITHUB.IO](#)

Introduction to Optimization and ... a month ago Title: Introduction to Optimization and Visualizing algorithms; Date: ...	Introduction to Optimization and ... a month ago Title: Introduction to Optimization and Visualizing algorithms; Date: ...	Introduction to Optimization and ... a month ago Title: Introduction to Optimization and Visualizing algorithms; Date: ...	Intro Optim a mont Title: I Optim algorit
---	---	---	--

Sponsored

Change Your Life Story in Canada!

Make Home Canada

[Learn More](#)

Aim to achieve your dreams with an SIP

Aim to achieve your dreams

Calcutta: The price (& size) of these hearing aids might surprise you

Hear.com

India: Why Everyone Is Excited Over This Rs.1999 Smartwatch

CrossFit

The Average IQ in Calcutta is 99. Check if your IQ is higher.

WW IQ Test

Škoda Slavia – Get Benefits of up to Rs.1.5 Lakh*

Škoda Auto India

[Book Now](#)

xavierbourretsicotte.github.io Comment Policy

All comments are welcome, feel free to contact me to discuss data, learning and maths

Got it

2 Comments

 1 Login ▾

G

Join the discussion...

LOG IN WITH

OR SIGN UP WITH DISQUS 

Name

 1

Share

Best Newest Oldest

A

Abdulkarim

2 years ago

great example. there is a slight mistake in the α matrix, last row should be $-1\alpha_2$ not $-2\alpha_2$

0 0 Reply • Share ›



Umer Farooq

3 years ago

You saved my day. Thanks

0 0 Reply • Share ›

Subscribe

Privacy

Do Not Sell My Data

Sponsored

Change Your Life Story in Canada!

Make Home Canada

[Learn More](#)

Aim to achieve your dreams with an SIP

Aim to achieve your dreams

Calcutta: The price (& size) of these hearing aids might surprise you

Hear.com

India: Why Everyone Is Excited Over This Rs.1999 Smartwatch

CrossFit

The Average IQ in Calcutta is 99. Check if your IQ is higher.

WW IQ Test

Airports Say Don't Use This Type Of Suitcase

investing.com

Navigation

[Data Blog \(\)](#)

[About me \(/pages/about-me.html\)](/pages/about-me.html)

[Categories \(/pages/categories.html\)](/pages/categories.html)

[atom \(/feeds/all.atom.xml\)](/feeds/all.atom.xml)

Author

[github \(https://github.com/xavierbourretsicotte\)](https://github.com/xavierbourretsicotte)

[linkedin \(https://www.linkedin.com/in/xavier-bourret-sicotte/\)](https://www.linkedin.com/in/xavier-bourret-sicotte/)

[stackexchange \(https://stats.stackexchange.com/users/192854/xavier-bourret-sicotte\)](https://stats.stackexchange.com/users/192854/xavier-bourret-sicotte)

Categories

[Machine Learning \(19\) \(/category/machine-learning.html\)](/category/machine-learning.html)

[Statistics \(4\) \(/category/statistics.html\)](/category/statistics.html)

Links

Centre National des Arts et Metiers (<http://www.cnam.fr/portail/conservatoire-national-des-arts-et-metiers-821166.kjsp>)

© Xavier Bourret Sicotte 2016

Powered by Pelican